

Pratyush Sudhakar

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EDUCATION

Cornell University

Ithaca, NY

Bachelor of Science in Computer Science (*Minors: Mathematics & Psychology*)

May 2026

Coursework: Adv. ML, Systems Prog., Algorithms (Java), Functional Prog. (OCaml), OOP (Java), Computer Vision

TECHNICAL SKILLS

Languages: Python, TypeScript/JavaScript, Java, C++, SQL, GraphQL, OCaml, HTML/CSS, Bash

Frameworks & Libraries: React.js, Next.js, React Native, Expo, Node.js, Express.js, FastAPI, Flask

Data, Cloud & Tools: PostgreSQL, SQLAlchemy, Redis, Docker, Kubernetes, AWS, GCP, Azure, Git, Terraform

EXPERIENCE

Full-Stack Developer, DECA Lab, Cornell University

January 2025 - Present

Tech: React Native, Expo, Next.js, FastAPI, SQLAlchemy, PostgreSQL, TypeScript

Ithaca, NY

- Built a full-stack audit platform with typed REST APIs spanning multi-role workflows—access control, draft saves, submission, and reporting—with i18n and accessibility configurations tailored for diverse field workers.
- Engineered an offline-first mobile app (Legend State/MMKV, background sync) enabling auditors to start, resume, and submit field audits reliably in low-connectivity environments across localized, accessible interfaces.
- Designed normalized scoring pipelines with client-side CSV/XLSX/PDF export; hardened draft-save flows against duplicate-key failures and extended data integrity guarantees across multi-round field testing.

Computer Systems Developer, Rizvi Lab, Cornell University

July 2024 - September 2024

Tech: Python, Linux/Bash, IoT, Embedded Systems, Git

Ithaca, NY

- Built a PyQt touchscreen interface for BeagleBone systems enabling real-time monitoring and remote control of food processing equipment; automated operations with shell scripts and cron jobs, reducing manual intervention by 60%.

Software Engineering Intern, rapStudy

June 2023 - August 2023

Tech: React.js, TypeScript, Redux, Firebase, Material UI, REST APIs

Los Angeles, CA

- Built a secure song-sharing system using React and 10+ custom Firestore rules for precise access control.
- Developed Redux-powered global audio-playback component, perfectly syncing with lyrics display for enhanced UX.

Data Engineering Intern, Cornell College of Engineering

June 2023 - September 2023

Tech: Python, Pandas/NumPy, Selenium, Next.js, TypeScript, ETL Pipelines

Ithaca, NY

- Engineered a scalable ETL pipeline processing financial data from 3,000+ companies via Selenium, NLP, and multithreading, reducing processing time by 83%.
- Built a live data dashboard using Next.js and Server-Sent Events, cutting latency by 66% with parallelization.

ACTIVITIES

Head of Engineering & President, Perfect Match: Online Matchmaking

January 2023 - May 2025

Tech: Next.js, TypeScript, Tailwind CSS, MongoDB, Redis, AWS, CI/CD

Ithaca, NY

- Directed 4 cross-functional teams through product launch, recruitment, onboarding, and business planning.
- Built analytics platform delivering real-time survey insights to 5,000+ users via cached Map-Reduce aggregations, reducing latency by 93%; automated personalized emails driving 600+ new active users.

Technical Product Manager, Cornell Digital Tech & Innovation

January 2023 - December 2023

Tech: React.js, DynamoDB, Cypress, Kubernetes, Docker, Flutter, Agile

Ithaca, NY

- Spearheaded agile development of cross-platform ride scheduling app with a team of 7 developers.
- Architected test-driven development pipeline using Cypress, Mocha, and Supertest (94% code coverage); orchestrated CI/CD with GitHub Actions and Kubernetes.

VENTURES & PERSONAL PROJECTS

Halo - Wearable AI | Swift (iOS), Python, OpenAI (LLMs), Redis, Celery, IoT

June 2025 - July 2025

Built a real-time face-recognition app (iOS+ESP32) with Apple Vision + OpenAI; refactored code (−40%), trimmed app size (−30%), and cut backend latency from 3 min to 50 s via a Celery/Redis redesign.

AI Tool (Millennium Sponsored) | Flask, Azure, RAG (Vector DB), LangChain

March 2024 - May 2024

Devised a course syllabus assistant with Azure-hosted REST API, LangChain, and Zilliz cluster for vector similarity search, achieving 50ms average response time for context-aware student queries.